MKM-PF-001 — Depth Sensitivity Analysis

Component: Property Flood Response — `MKM-PF-001` **Scope:** How the priced damage ratio responds to error in the model's single most material uncertain input — the estimated flood **depth** at a property — across both the intensive margin (given a flood, how much) and the extensive margin (whether the property floods at all). **Date:** 2026-07-30. **Method:** closed-form, data-free; composes the model's own calibrated depth-damage curve (`config.damage.DD_POLY_COEFFS`) via `models.floodrisk.sensitivity`.

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1. Why depth is the material input

Everything the model prices flows from the effective depth `d = max(0, depth - floor_level - stilt)` through the depth-damage curve `g(d) = sum_i c_i * d^i`. The depth itself is the least certain quantity in that chain: the water-surface elevation is interpolated from a sparse gauge network by inverse-distance weighting (IDW) and then differenced against a DEM ground level, so **both an interpolation bias and a DEM error land directly on d**. The curve coefficients are calibrated and comparatively stable; the depth is not. This analysis therefore holds the curve fixed and perturbs the depth.

2. Intensive margin — depth error is attenuated

Given that a property floods, the damage ratio is a **concave, saturating** function of depth, so its elasticity `E(d) = d*g'(d)/g(d)` is below one everywhere and falls as water deepens:

depth d (m)	0.1	0.5	1.0	2.0	4.0
elasticity E(d)	0.96	0.79	0.64	0.52	0.49

A representative 1.0 m flood sits at a 40.2% damage ratio. Propagating a plus/minus 40% depth band through the curve gives:

percentile	depth (m)	damage	rel. to median	passthrough
p05	0.60	0.281	0.70	0.75
p50	1.00	0.402	1.00	—
p95	1.40	0.492	1.22	0.56

A 40% depth error becomes roughly a -30% / +22% damage error — **damped, not amplified**. Intensive-margin loss is robust to modest depth error, and the robustness improves in deep water where the curve flattens toward total loss.

3. Extensive margin — incidence is the real exposure

The attenuation above assumes the property floods. Whether it floods is governed by the same depth against the floor+BRI threshold, and there the response is a **step**. Take a property whose water stands 0.15 m over its threshold (damage 0.085):

depth bias (m)	effective depth	floods?	damage
-0.30	-0.15	no	0.000
-0.15	0.00	no	0.000
0.00	+0.15	yes	0.085
+0.15	+0.30	yes	0.159

depth bias (m)	effective depth	floods?	damage

A -0.15 m depth bias — well inside DEM/IDW error — moves this property from a positive damage ratio to **zero**. Aggregated over a portfolio, a small *systematic* depth bias therefore swings the **flooded count**, not just each property's severity. This is exactly the documented inverse-distance dilution failure mode, in which IDW smoothing pushed most properties just below threshold and left 178 of 200 unflooded. The marginal damage per metre `g'(d)` is largest at the toe (0.55/m near the threshold vs 0.20/m at 1.4 m), which is why the incidence margin dominates the model's loss uncertainty.

4. What this does and does not establish

The elasticities and the incidence step are exact properties of the calibrated curve — they are reproducible and not a Monte-Carlo estimate. What they do **not** establish is the accuracy of the depth estimate itself: on a synthetic catchment the interpolation recovers an assumed water surface, so this analysis bounds the *consequence* of a depth error, not its *size*. Sizing the depth error needs real gauge and survey data (validation question VQ-005, data quality — outstanding). The practical implication is a control priority: the model is most exposed at the flood/no-flood boundary, so validation effort belongs on the interpolation and DEM at threshold, not on the damage curve's shape.

Reproduce: `models.floodrisk.sensitivity` — `depth_damage_sensitivity`, `depth_bias_sensitivity`, `damage_distribution`; figures from the calibrated `DD_POLY_COEFFS` of 2026-07-30.